Design and Development of low-cost Myoelectric Prosthetic Arm for Upper Limb Amputees

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Abstract— **This paper addresses the prevalent challenges in the field of prosthetic devices, such as comfort, use cases, cost efficiency, customization, and technical hurdles. In this work, a functional, 3D-printed Myoelectric Prosthetic Arm prototype developed using electromyographic (EMG) pattern recognition to help amputees of upper limbs. With flexible joints that ensure durability and improve impact resistance, EMG sensors are incorporated into the design to enable precise hand functions in this open-source, low-cost Myoelectric Arm which also enable motor control and EMG pattern recognition. Its affordability also makes it more useful for people in developing countries who have an amputation. This research shows that an integration of EMG Pattern and 3D Printing enhances an ability to grasp delicate objects, which distinguishes this prosthetic arm from conventional myoelectric prostheses that do not have these functionalities. Long-term testing of the hand's functional capabilities is also made easier by its smooth integration into standard sockets. This research is contributing the prosthetic devices to make them more affordable, accessible, and technologically advanced solution for amputees of upper limbs.**

Introduction

Addressing the problem of designing a low-cost, convenient, reliable, and efficient prosthetic hand for disabled individuals who are unable to perform daily activities due to arm amputations or musculoskeletal disorders. Four types of upper limb prosthetics currently exist: cosmetic, mechanical, myoelectric, and hybrid. Efforts have been made to implement control algorithms based on electromyography (EMG) signals to control upper limb prostheses using the MG995 Servo motors and 3D Printing Stack. EMG signals provide non-invasive access to muscle contraction processes and can be used to control prosthetics by Servos. Commercial devices such as the Michelangelo hand, i-Limb Ultra hand, and Be-bionic 3 hand are available but are expensive for most disabled people in developing countries. Previous studies have explored cheap upper limb prosthesis designs, but they have limitations such as limited functionality or lack of grip characteristic data. The paper presents a new approach to designing an artificial arm with EMG multi finger control with the help of 3D printing technology, Servos and Teensy 4.0 as a microcontroller, capable of performing commonly used gripping actions with approximate force and grip.

I. SETUP DESCRIPTION

A prosthetic hand prototype consists of two primary components:

A. Mechanical Unit

The Mechanical Portion of the Myoelectric Hand is built using 3D Printing. Following the tensile strength test, the decision was made to print an arm out of PLA. It enables us to obtain.
(1) Tensile strength 8.00 - 103 MPa,
(2) Bending strength 43.6–59.6 MPa,
(3) Compressive strength 48.2–62.0 MPa,
(4) Self-ignition temperature 388°C.
The hand is a complex joint that joins the arm to the body of an amputee. There are four fingers index, middle, ring, and pinky, with a separation of 3 joints i.e., distal, middle, and proximal

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and a thumb with 2 separations of two joints i.e. Metacarpal bones. Rubbers are ties on these joints to make it flexible and elastic. So as after successful movement of finger, it should go to their initial place when motor moves to home position. The Fingers are connected to the 3D Printed Palm so as to hold the grip as a natural human hand. Each finger can be represented as a serial manipulator, where tip of the finger is tied with the thread and passed from the distal, middle, and proximal parts of single finger to the servo motor, which can act as a power transmission from servos to the finger. In current research we use the mechanism in a simplified form. Namely, there is a hinge mechanism between the distal, middle, and proximal phalanxes. These three phalanxes are attached statically to each other with a supporting pin by which the joints can move freely. Based on above simplifications, the joint mechanism of the finger was designed and assembled. A 3D schematic model is presented fig 1. For the Arm, it's also 3D Printed and is a cavity like structure designed in such a way that it can hold all the electronics part including Microcontroller and circuits in it. The end of Arm can be connected to the upper limb amputees using well-crow for proper fixture.

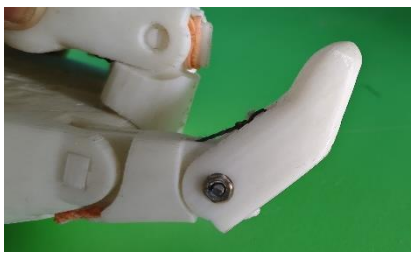

Figure 1. Mechanical Finger

Finger	Action (G)		
	1	2	3
Thumb (Distal phalanx)	0°	30°	60°
Thumb (Carpo-metacarpal joint)	0°	20°	50°
Index	0°	45°	110°
Middle	0°	45°	110°
Ring	0°	35°	90°
Minimus	0°	35°	85°

Figure 2. Dependence of servo angle with respect to finger position.

B. Electronic Hardware unit

The electronic hardware unit of the prosthetic hand is controlled by an Arduino Nano, which is a miniature board on which Arduino Nano is soldered/fixed with an external DC to DC voltage buck converter for power Amplification. The EMG Sensor and MG995 Servos can be easily connected directly to the pins of Nano with all other sensors integrated in the hand. It has 8 analog inputs and 14 digital inputs, with each analog input capable of being used as a digital output. It also has PWM outputs and transistors for controlling servo speed, LED brightness, and power of heaters. The unit utilizes EMG sensors from DF Robotics for muscle activity detection, which amplify the minimum electron diffraction signal and suppress noise. The sensors require a stabilized DC voltage and are placed on the upper and lower parts of the forearm in a specific order.

The Electronic Hardware Unit consists of following components.

i. Arduino Nano

There are eight analogue inputs and fourteen digital inputs/outputs on the Arduino Nano. Most important, every analogue input has the ability to be used as a digital output. In addition, two of the digital inputs function as I2C communication channels, three are used for SPI communication, and six of the digital inputs can function as pulse width modulators (PWM). The Arduino ISP pad, which provides a port for flashing the microcontroller's firmware, is located on the other end of the board, away from the micro-USB connector. PWM outputs and integrated transistors are essential for managing a number of parameters, including servo speed, LED brightness, and heater power levels, among other things. The presence of analog inputs facilitates the reading of data from analog sensors, adding versatility to the board's capabilities. Additionally, Digital 2 and Digital 3 are assignable for use as external interrupts. These external interrupts are essential for alerting the microcontroller to significant events. They trigger specific interrupt processing routines, enabling actions like exiting power-saving modes and performing necessary calculations.

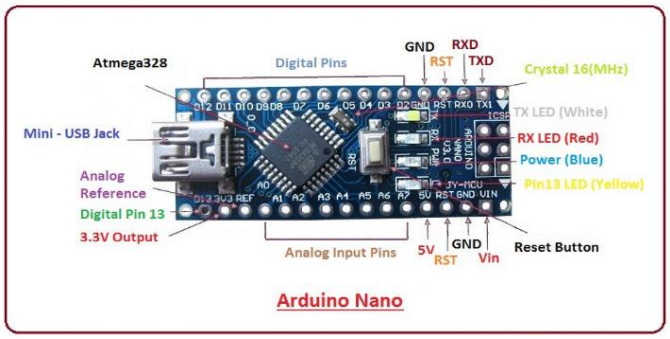

Figure 3. Arduino Nano

ii. EMG Muscle Sensor

EMG sensors, also known as electromyography sensors, measure the electrical signals generated by muscles when they move. These sensors are based on the fact that when muscles contract, they release an electrical impulse that circulates through adjoining bone and tissue. EMG sensors amplify and process the complex electrical activity of a muscle and convert it into a simple analog signal. This signal can be read by any microcontroller with an analog-to-digital converter (ADC). EMG sensors are used to: Record muscle movement, measure small electrical signals generated by muscles when they move, and help detect neuromuscular abnormalities. EMG sensors can measure movements such as: Lifting your arm up, clenching your fist, and moving a finger.

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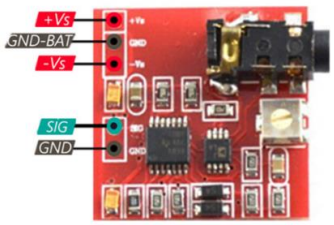

Figure 4. EMG Muscle Sensor

iii. MG995 Tower-Pro Servo Motor

MG995 Metal Gear Servo Motor is a high-speed standard servo can rotate approximately 180 degrees (60 in each direction) used for airplane, helicopter, RC-cars and many RC model. Provides 10kg/cm at 4.8V, and 12kgcm at 6V. It is a Digital Servo Motor which receives and processes PWM signal faster and better. It equips sophisticated internal circuitry that provides good torque, holding power, and faster updates in response to external forces. They are packed within a tight sturdy plastic case which makes them water and dust resistant which is a very useful feature in RC planes, Boats, and RC Monster Trucks etc. It equips 3-wire JR servo plug which is compatible with Futaba connectors too.

Figure 5. MG995 Servo Motor

Specifications:
Weight : 55g
Dimension : 40.7 × 19.7 × 42.9 mm
Operating Speed (4.8V no load): 20sec / 60 deg
Operating Speed (6.0V no load): 16sec / 60 deg (no load)
Stall Torque (4.8V): 10kg-cm
Stall Torque (6.0V): 12kg-cm
Operation Voltage : 4.8 - 7.2Volts
Gear Type: All Metal Gears
Stable and shock proof double ball bearing design
Dead band width: 5 µs
Temperature range: 0 ºC – 55 ºC.
Control System: Analog
Operating Angle: 120degree
Required Pulse: 900us-2100us.

iv. LM2596 DC-DC Buck Converter

A DC-to-DC converter that increases current from its input (supply) to its output (load) while decreasing voltage is known as a buck converter or step-down converter. It belongs to the switched-mode power supply class. As DC-to-DC converters, switching converters (like buck converters) offer significantly higher power efficiency than linear regulators, which are more straightforward circuits that step up the output current but do not step down the voltage by dissipating energy as heat(Source). Buck converters are helpful for tasks like converting a computer's main supply voltage, which is typically 12 V, down to lower voltages needed by USB, DRAM, and the CPU, which are typically 5, 3.3, or 1.8 V. Buck converter efficiency can be very high, often over 90%.A diode and a transistor are the two semiconductors that most buck converters have in common, though more recent models usually include a second transistor for synchronous rectification in place of the diode, as well as at least one energy storage component (an inductor, capacitor, or both combined). Filters consisting of capacitors, sometimes in conjunction with inductors, are typically added to the input and output of such a converter (supply-side filter) to reduce voltage ripple.[/2] The inductor that "bucks," or resists, the supply voltage is the source of its name.[/3] The typical switching frequency range for buck converters is between 100 kHz and a few MHz. Smaller inductors and capacitors can be used with a higher switching frequency, but this comes at the cost of increased lost efficiency from more frequent transistor switching.

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II. METHODOLOGY

Figure 6. Design Method

A. *Customized Dimensions and Design*

i. 3D Printing and Measurement: Methods for getting accurate limb measurements from the user, as Precise measurements are necessary for a perfect fit.

ii. Computer Aided Design (CAD): How custom prosthetic designs are made using CAD software, parameters like the distribution of weight and the choice of material.

iii. Technology and Materials: An overview of materials that are frequently used in prosthetic hand 3D printing. technologies used in 3D printing.

iv. The Printing Method Detailed description of the 3D printing procedure. Durability or structural integrity are considered.

B. *Manual Operation*

i. Articulation and grasping: The manual ability of the 3D-printed hand is designed for it. Finger and joint mechanisms. Interface and Attachment link to the user's remaining appendage. Mechanisms of attachment, sockets, and straps.

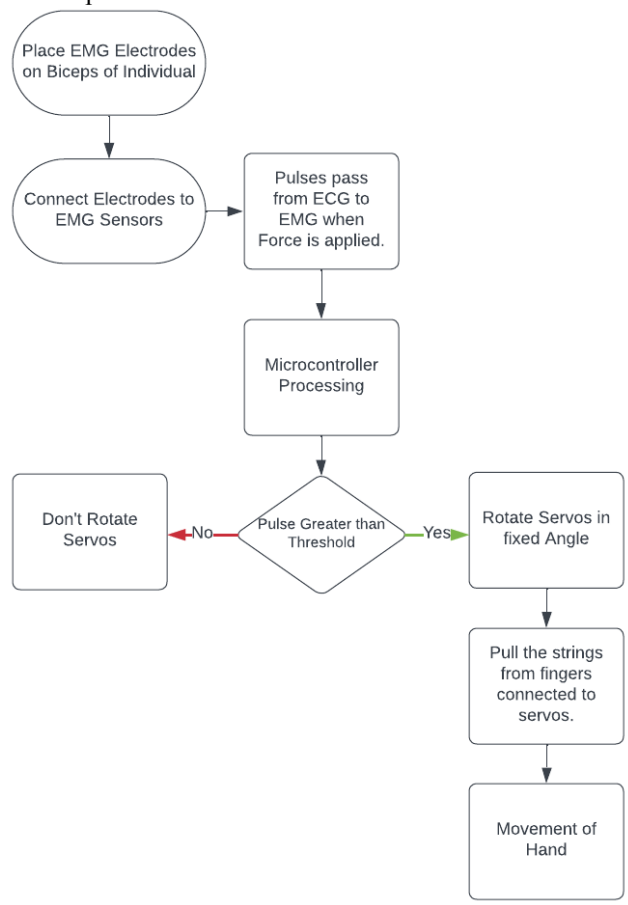

Figure 7. Process Flow Chart.

ii. Integrated Circuits: Technologies of Sensing Overview of sensors, including force, accelerometer, and EMG sensors. How sensors interpret muscle movements and user intent.

iii. Unit of Control: An overview of the processing unit or microcontroller that interprets sensor data. Signal processing and embedded software's function.

iv. Motors and Actuation: How the prosthetic fingers move is managed by motors. Haptic feedback integration for sensory perception. Putting together the detailed instructions for assembling the electronics and 3D-printed hand. Combining motors, sensors, and control units.

C. *Testing and calibration*

i. Sensor Calibration: The need for precise sensor calibration to ensure accurate control.

ii. Functional Testing: The importance of functional testing to guarantee proper performance. Real-world testing with end-users. Results and Discussion Presentation of results

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from testing. User feedback and experiences. Discussion of the advantages and limitations of the proposed system.

III. RESULTS ANALYSIS

We look at the use of EMG sensors for prosthetic hand force control. The corresponding fingers are meant to contract to move that finger. We take into consideration three finger positions in order to execute this type of movement. That is, an entirely bent, partially bent, and straight finger. Such a choice is driven by fundamental requirements for carrying out grasping actions. Moreover, multifinger control facilitates organic grasping of objects. An algorithm for identifying muscle activity is explained in (Unanyan & Belov, 2020). Its computing complexity is low, its training mode is quick and simple, and its accuracy is sufficient. Real-time EMG data processing during the separate bending of each finger is shown in Fig. 10. Here, G stands for the kind of action that the mechanical arm will carry out.

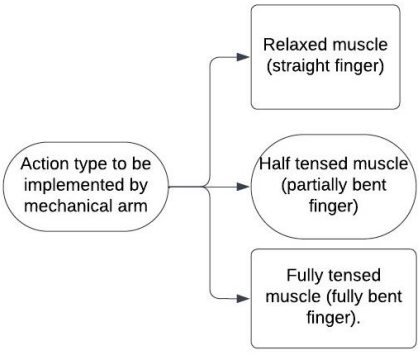

Figure 8. Action Type

A. Threshold Clarification

The Arduino program includes a preset threshold value, which is set to 100, to regulate the prosthetic hand's control mechanism. The electrocardiogram (ECG) electrodes detect electrical signals produced by an amputee's muscular contractions, which are then provided to the electromyographic (EMG) sensors. The signals from these EMG sensors are then sent to the Arduino for further signal processing. The Arduino Integrated Development Environment (IDE)'s serial monitor can be used to track and view the threshold values, which are received as pulses.

After the Arduino Nano records the analogue values resulting from the muscle contractions, they are converted from analogue to digital format. The digital values are then contrasted with the preset threshold value of 100. The recorded values act as a trigger, sending a command to the prosthetic hand's motors if they are higher than this threshold. The prosthetic hand can then move and function with controlled movement because these motors actuate at specific angles in response to commands. The user can precisely manipulate and control the prosthetic hand's movements in response to their own muscle signals thanks to this dynamic process.

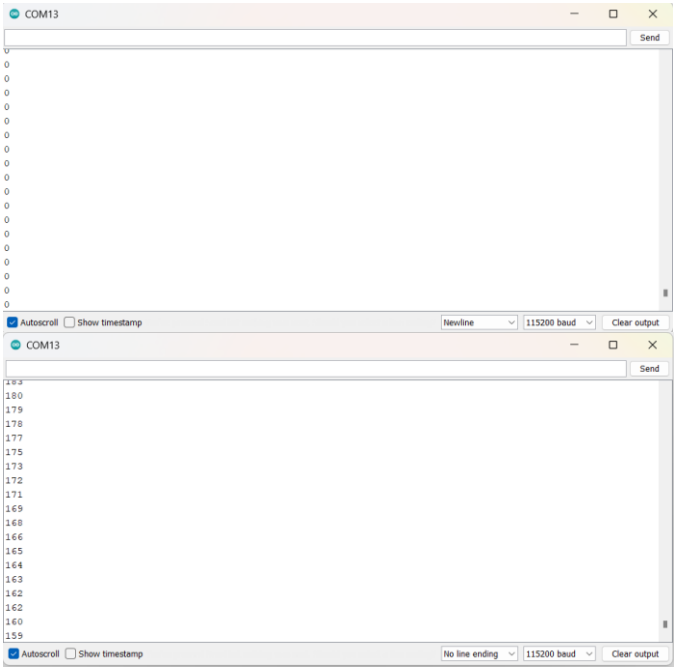

Figure 9.Threshold Values on Serial Monitor.

CONCLUSION

In conclusion, this study addressed the primary problems related to prosthetic devices by introducing an innovative and creative solution: the prototype of a 3D-printed myoelectric prosthetic arm. Our research has addressed problems with comfort, affordability, customization, and technical difficulties, providing a viable path to improve the quality of life for people who have had upper limb amputations or musculoskeletal conditions. Through the integration of electromyographic (EMG) pattern recognition, we have produced a prosthetic arm that is both affordable and highly functional, making it an impressive piece of technology. Flexible joints have been added, which has increased durability and impact resistance while enabling precise hand functions that allow users to grasp delicate objects. Our prosthetic arm differs from traditional myoelectric prostheses with this feature.

In addition to providing a practical solution to the issue of creating effective and affordable prosthetic hands, this research sheds light on important prosthetic design components, such as precise measurement, CAD design, material selection, and 3D printing. The prosthetic hand's manual function, which includes grasping and articulation, shows how well it can replicate natural hand movements. In conclusion, this research marks a substantial advancement in the field of prosthetic devices with the goal of lowering the cost and increasing the accessibility of sophisticated and useful solutions for those who have lost an upper limb. EMG pattern recognition combined with 3D printing technology has the potential to change people's lives by providing them with a more practical, dependable, and effective way to regain function in their limbs.

ACKNOWLEDGMENT

Our heartfelt appreciation goes out to everyone who helped make the 3D-printed Myoelectric Prosthetic Arm prototype and

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the research described in this paper a reality. Above all, we would like to express our gratitude to the amputees and people with musculoskeletal disorders who so kindly took part in the testing and feedback procedure. Their insightful comments and invaluable insights have been crucial in helping us customise this prosthetic arm to suit their preferences and needs. We sincerely appreciate all of the advice, mentoring, and help that our advisors and mentors have given us during this research project. Their knowledge and support have been invaluable in guaranteeing the calibre and inventiveness of our work..

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